A Hands-On Evaluation of UniverSeg for Few-Shot Medical Image Segmentation

Author: JungJu Jo[1,2]

Course Instructor: Dr. Donghoon Kang Adjunct Professor

(Fundamentals of Deep Learning and PyTorch Programming)

[1]*Radiological and Medico-Oncological Sciences, University of Science and Technology, Seoul, Republic of Korea*

[2]*Radiation Therapy Technology and Standards, Korea Institute of Radiological and Medical Sciences, Seoul, Republic of Korea*

[3]Uni*versity of Science and Technology (UST) Senior Researcher, Intelligence and Interaction Research Center, Korea Institute of Science and Technology (KIST)*

Abstract

This study presents a practical evaluation of UniverSeg, a pretrained universal model for few-shot medical image segmentation. The model was applied without any additional training to white blood cell microscopy images and brain MRI data using a small number of support image–label pairs. We investigated the effects of support label switching, support set size, input intensity inversion, and ensemble inference on segmentation performance. Experimental results demonstrate that UniverSeg can adapt to different segmentation targets and imaging domains through support-conditioned inference, achieving stable performance in data-limited scenarios.

1. Introduction

Medical image segmentation is a fundamental technique in clinical applications such as computer-aided diagnosis, treatment planning, and disease progression analysis. Although deep learning–based methods have achieved remarkable performance, they typically require large-scale annotated datasets. In medical imaging, however, collecting pixel-level annotations is costly and time-consuming, particularly in scenarios involving rare diseases, new imaging modalities, or institution-specific acquisition protocols

To address these limitations, few-shot and universal segmentation approaches have recently attracted increasing attention. Few-shot segmentation aims to perform new segmentation tasks using only a small number of annotated examples, making it well suited for data-limited medical applications. Among these approaches, UniverSeg (Universal Medical Image Segmentation), introduced at ICCV 2023, proposes a universal framework that enables segmentation of diverse medical images without additional training or fine-tuning. By adopting an in-context learning paradigm, segmentation tasks are defined at inference time through a small set of support image–label pairs rather than through task-specific retraining

Figure 1 illustrates the overall architecture of UniverSeg. The model follows an encoder–decoder structure similar to U-Net, but incorporates a support-conditioned inference mechanism through cross-block connections. The query image and support images are processed in parallel pathways, and feature representations from the support branch are integrated into the query branch at multiple resolution levels. This design allows the model to adapt its segmentation behavior based on the provided support examples, enabling flexible segmentation across different targets and imaging domains.

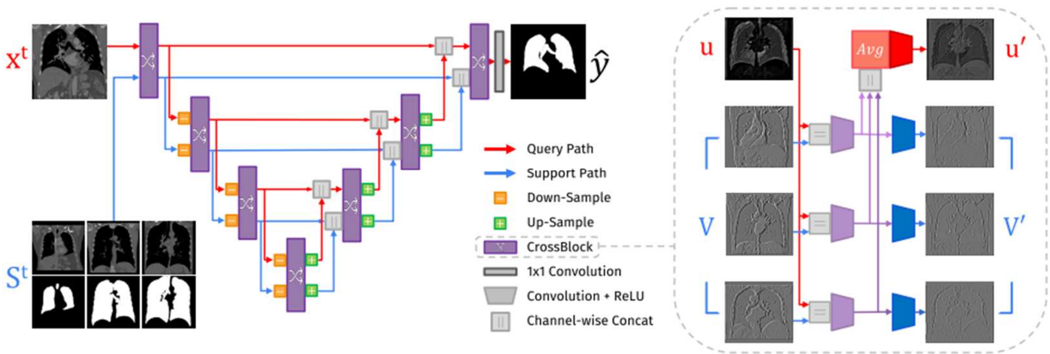

Fig. 1. A UniverSeg network (left) takes as input a query image and a support set of image and label-maps (pairwise concatenated in the channel dimension) and employs multi-scale CrossBlock features. A CrossBlock (right) takes as input representations of the query u and support set V = {vi}, and interacts u with each support entry vi to produce u′ and V ′.

The objective of this project is not to propose a new segmentation method, but to evaluate and understand the behavior of a pretrained UniverSeg model on real medical imaging datasets. Through practical experiments, we analyze how the model adapts to different segmentation targets, support set configurations, and input variations, and assess its applicability and limitations in few-shot medical image segmentation.

2. Materials and Methods

2.1 Experimental Environment

All experiments were conducted using a Python-based deep learning framework. The experimental environment is summarized as follows.

Development environment: Visual Studio Code
Programming language: Python 3.10.18
Deep learning framework: PyTorch
GPU: NVIDIA GeForce RTX 5070
CUDA: CUDA-enabled PyTorch (CUDA 12.8)
PyTorch version: 2.10.0

All inference procedures were performed on the GPU when available. No model training or fine-tuning was conducted during the experiments.

2.2 Model Overview

In this study, we employed UniverSeg (Universal Medical Image Segmentation), a pretrained global segmentation model designed for few-shot medical image segmentation. UniverSeg enables segmentation of previously unseen tasks without any additional training by conditioning the inference process on a small set of example annotations.

The model takes the following inputs:

Query image: the target image to be segmented

Support set: a small collection of images paired with corresponding segmentation masks that define the segmentation task

Given a query image x and a support set $\mathcal{T} = \{(x_n, y_n)\}$, the model predicts the segmentation mask y as

$$y = f(x, \mathcal{T})$$

where y denotes the predicted segmentation output. In this formulation, the support set serves as task-defining contextual information rather than training data.

2.3 Datasets

2.3.1 White Blood Cell (WBC) Dataset

To evaluate the performance of UniverSeg, the White Blood Cell (WBC) segmentation dataset was used. This dataset consists of microscopy images of white blood cells with pixel-wise annotations.

Two segmentation labels were considered in this study: Cytoplasm, Nucleus

The dataset was divided into two subsets: Support set: used to define the segmentation task

Test set: used for performance evaluation

All images are grayscale and were normalized to a pixel intensity range of [0, 1].

2.3.2 OASIS Brain MRI Dataset

To further assess domain generalization, the OASIS brain MRI dataset was utilized. This dataset contains mid-coronal slices of brain MRI scans with corresponding binary segmentation labels for anatomical structures. The OASIS dataset was used exclusively for qualitative evaluation.

2.4 Support Set Construction

The support set is a critical component in UniverSeg, as it defines the segmentation task at inference time. For each experiment, support image–mask pairs were randomly sampled from the designated support split.

The following support set sizes were evaluated: $N = 1,2,4,8,16,32,64$

Each support image was paired with its corresponding segmentation mask using identical indices. All support images underwent the same preprocessing steps as the query images.

2.5 Inference Procedure

All segmentation results were obtained using the pretrained UniverSeg model without any additional training. Query images, support images, and support masks were jointly provided as inputs to the model during inference.

The model outputs logits, which were transformed into probability maps using a sigmoid activation function. Final binary segmentation masks were generated by applying a threshold of 0.5.

2.6 Evaluation Metric

Segmentation performance was evaluated using the Dice Similarity Coefficient (DSC), defined as

$$\mathrm{DSC} = \frac{2 \mid A \cap B \mid}{\mid A \mid + \mid B \mid}$$

where Arepresents the predicted segmentation mask and Bdenotes the ground truth mask. Dice values range from 0 to 1, with higher values indicating better segmentation performance.

2.7 Additional Experimental Settings

2.7.1 Support Label Switching

To evaluate task adaptability, the segmentation label in the support set was changed between cytoplasm and nucleus while keeping the model parameters fixed.

2.7.2 Input Image Transformation

Intensity inversion was applied to both support and query images to assess robustness to appearance changes.

2.7.3 Effect of Support Set Size

The effect of varying the support set size on segmentation performance was systematically analyzed.

2.7.4 Ensemble Inference

Multiple predictions were generated using different support sets, and the final output was obtained by averaging the soft prediction maps.

2.8 Implementation Details

All inference procedures were executed under a torch.no_grad() context to avoid unnecessary gradient computation. Visualization was performed using Matplotlib, and tensors were converted using detach() and cpu() operations prior to rendering.

3. Results

3.1 WBC Cytoplasm Segmentation Performance

The pretrained UniverSeg model was first evaluated on the cytoplasm segmentation task using the White Blood Cell (WBC) dataset. For this experiment, a support set was constructed from the designated support split, and inference was performed on randomly selected samples from the test split without any additional training or fine-tuning.

Quantitative evaluation using the Dice Similarity Coefficient (DSC) on 10 test images showed Dice scores ranging from 72.9% to 96.1%, with a mean Dice of 88.61 ± 7.95% (mean ± SD). Visual inspection of representative examples (Fig. 2) indicates that the predicted masks closely follow the cytoplasm boundaries in most cases, capturing both the overall cell morphology and internal concavities. Minor discrepancies were mainly observed near ambiguous boundaries or in low-contrast regions.

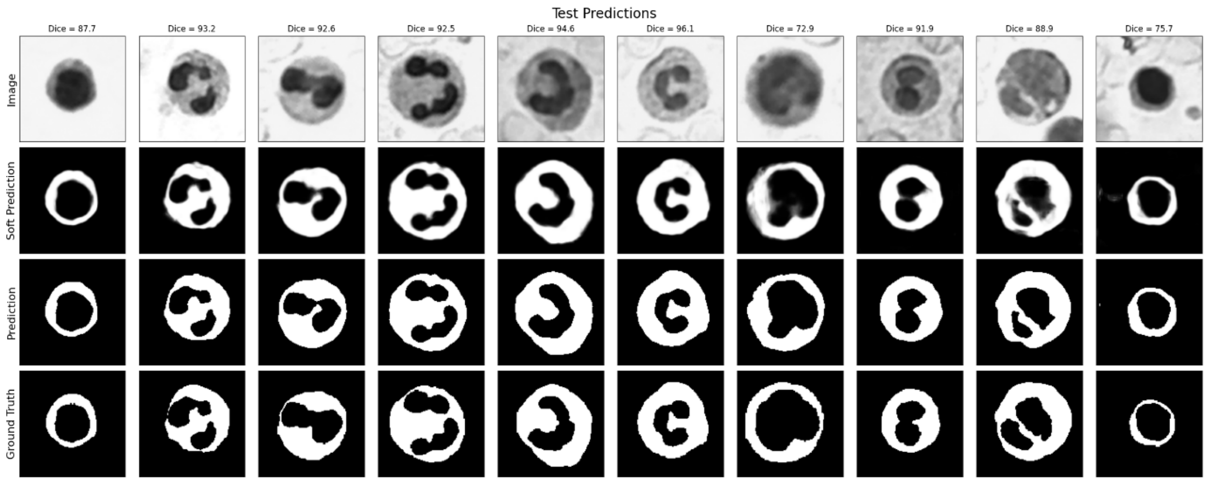

Fig. 2. Representative WBC cytoplasm segmentation results using UniverSeg. For each example, the input image, soft prediction, hard prediction, and ground truth mask are shown.

3.2 Task Switching via Support Label Change: Nucleus Segmentation

To assess the flexibility of UniverSeg in adapting to different segmentation targets, the support masks were changed from cytoplasm to nucleus labels while keeping the model parameters fixed. This experiment evaluates whether the segmentation task can be redefined solely through the support set.

On 10 test samples, nucleus segmentation achieved Dice scores between 75.6% and 97.7%, with a mean Dice of 92.99 ± 6.58%. As illustrated in Fig. 3, the predicted masks clearly shift their focus from cytoplasm regions to nuclear structures, demonstrating that the model effectively interprets the support masks as task-defining conditions rather than relying on fixed label semantics.

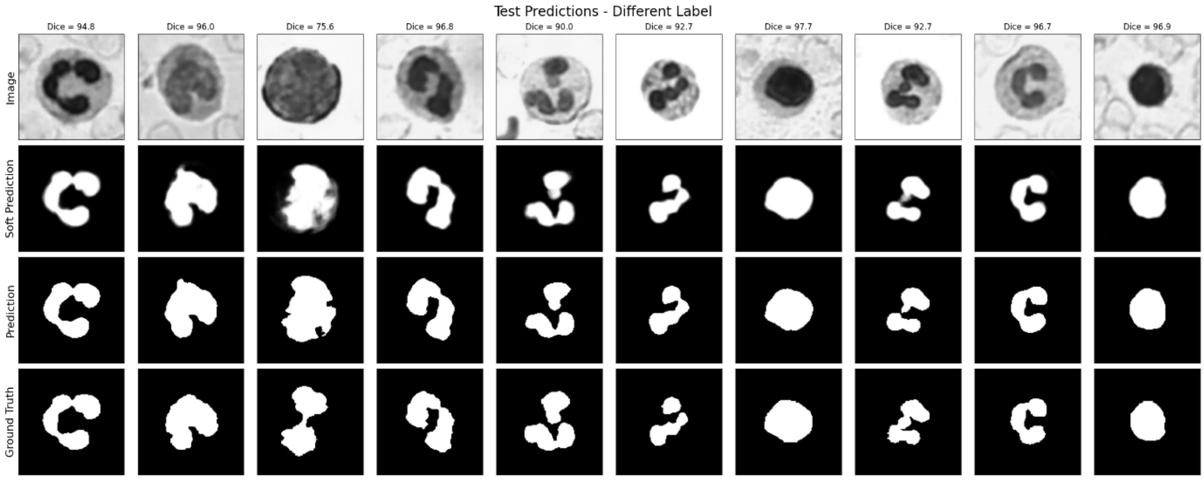

Fig. 3. Qualitative results of WBC nucleus segmentation obtained by changing only the support labels, demonstrating task switching without retraining.

3.3 Robustness to Intensity Inversion

Robustness to input appearance changes was evaluated by applying intensity inversion (1 − image) to both support and query images for the cytoplasm segmentation task. This experiment tests whether UniverSeg maintains performance under domain shifts in pixel intensity.

Despite the altered intensity distribution, the model preserved segmentation performance, yielding Dice scores between 70.7% and 95.5%, with an average of 87.62 ± 8.30% across 10 test samples. As shown in Fig. 4, the predicted masks remain structurally consistent with the ground truth, indicating that UniverSeg relies on support-conditioned structural cues rather than absolute intensity patterns.

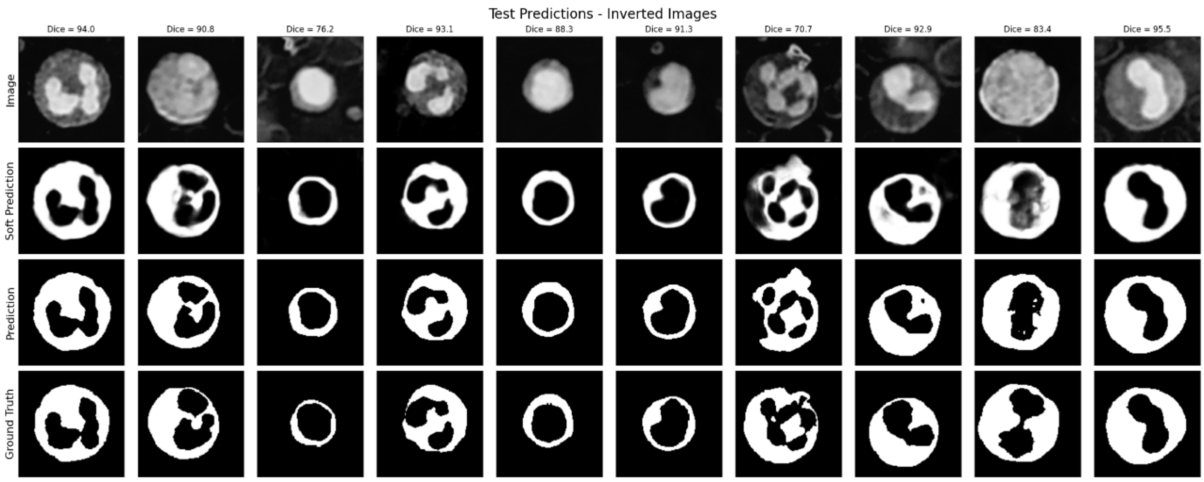

Fig. 4. Cytoplasm segmentation results on intensity-inverted WBC images, illustrating robustness to appearance changes.

3.4 Effect of Support Set Size

The impact of support set size on segmentation performance was investigated by varying the number of support examples $N = 1,2,4,8,16,32,64$. Predictions were generated for the same query images, and Dice scores were averaged over three representative test samples.

Results show a clear trend of performance improvement as the support set size increases. Very small support sets ($N = 1,2$) resulted in unstable and lower Dice scores, while substantial gains were observed when increasing the support size to $N = 4$and $N = 8$. Performance improvements became more gradual beyond $N = 16$, indicating diminishing returns at larger support sizes. Qualitative comparisons in Fig. 5 reveal progressively refined boundaries and reduced segmentation artifacts with increasing N.

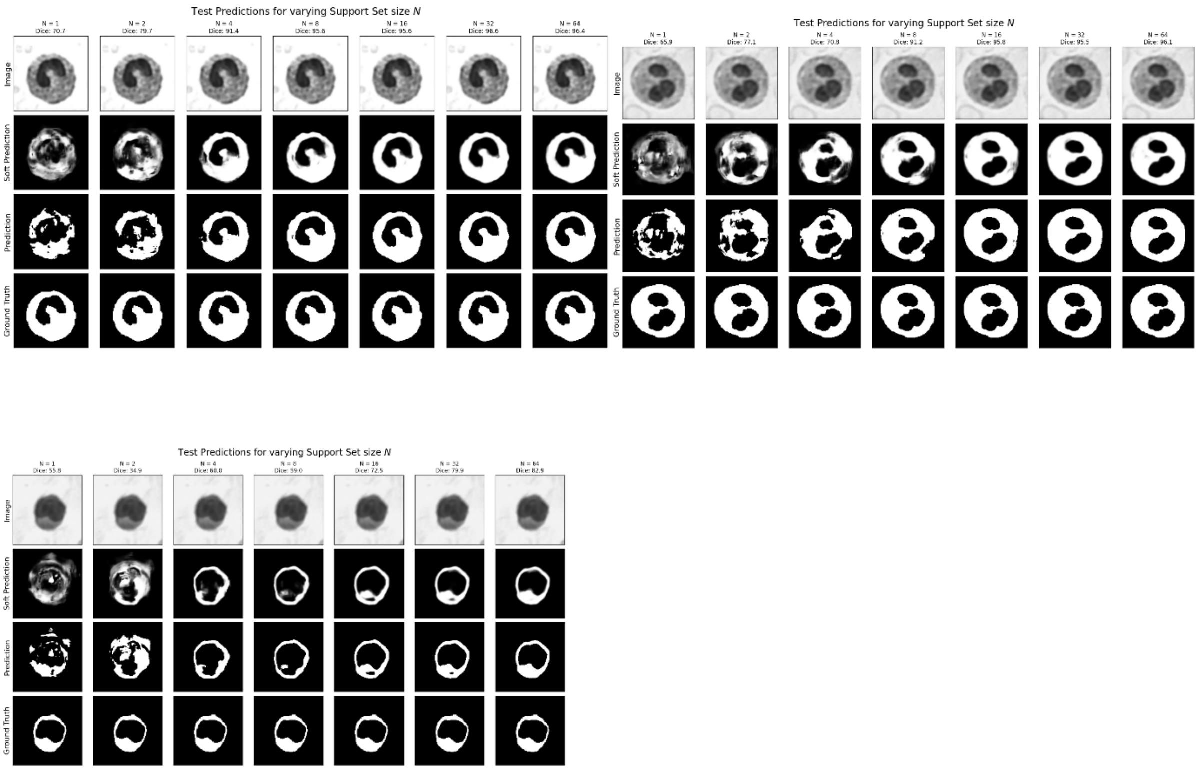

Fig. 5. Segmentation results for varying support set sizes, showing improved accuracy and stability as the number of support examples increases.

3.5 Multi-label Segmentation

Although UniverSeg employs a binary segmentation architecture, multi-label segmentation was achieved by performing independent inference for each label and combining the results. This approach was applied to WBC images containing multiple anatomical structures.

Across 10 test images, the multi-label segmentation achieved Dice scores ranging from 75.0% to 94.9%, with a mean Dice of 87.97 ± 6.16%. Visual results (Fig. 6) demonstrate that multiple structures can be segmented simultaneously, although minor competition between adjacent regions was occasionally observed near boundaries.

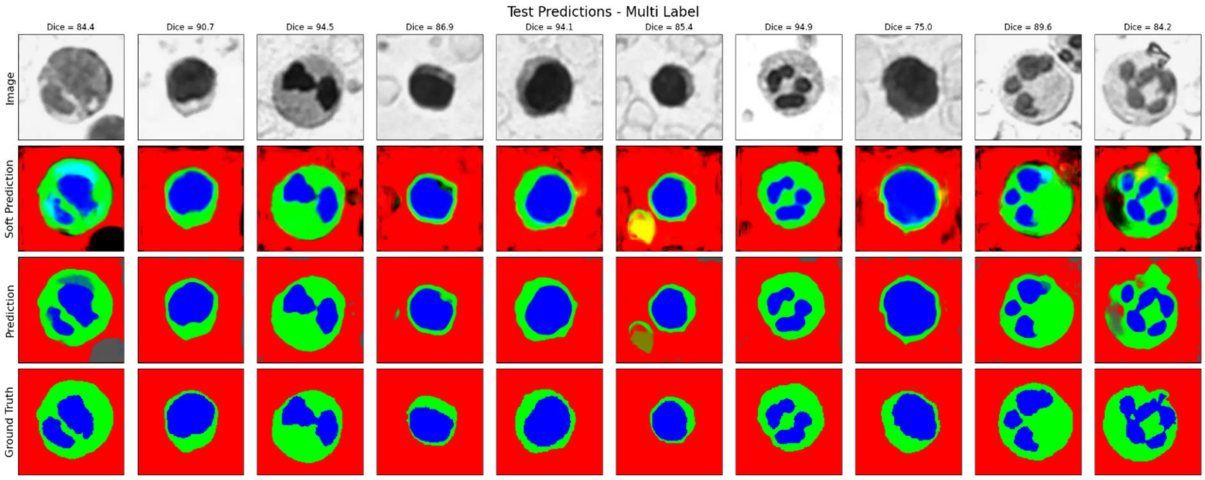

Fig. 6. Examples of multi-label segmentation results obtained by combining label-wise binary predictions.

3.6 Effect of Ensembling

To reduce prediction variability caused by random support set sampling, an ensemble strategy was applied. With the support set size fixed at $N = 8$, five different support sets were randomly sampled, and predictions were generated independently. The final ensemble prediction was obtained by averaging the soft prediction maps.

For three representative test images, ensemble Dice scores consistently exceeded the mean Dice of individual predictions. Specifically, improvements of approximately +2.66%, +1.78%, and +3.80% were observed compared to the corresponding single-prediction averages. As illustrated in Fig. 7, ensembling reduced localized false positives and false negatives, leading to smoother and more consistent segmentation boundaries.

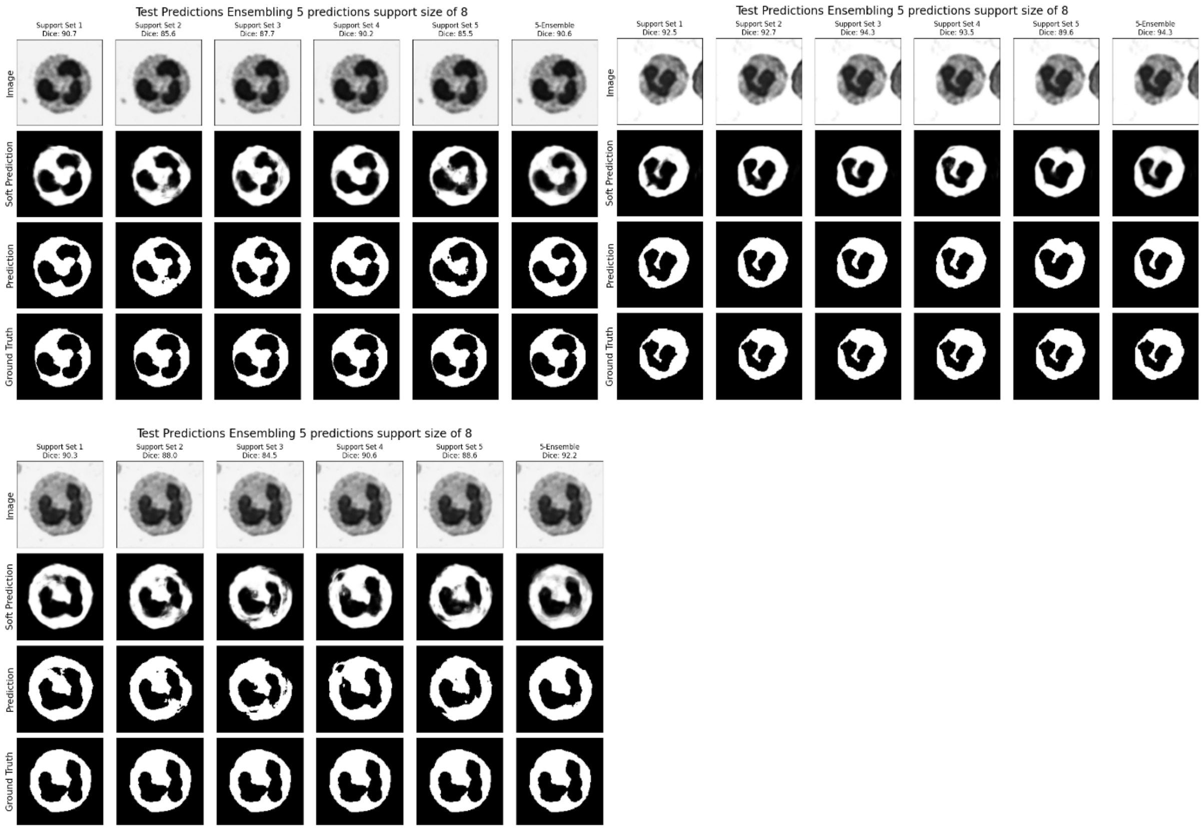

Fig. 7. Comparison of individual predictions using different support sets and the corresponding 5-model ensemble result, demonstrating improved segmentation consistency.

3.7 Qualitative Results on the OASIS Brain MRI Dataset

To evaluate domain generalization, UniverSeg was applied to the OASIS brain MRI dataset using a support set of 10 images. Although quantitative Dice scores were not computed for this experiment, qualitative results indicate that the model successfully captures the target brain structures in mid-coronal slices.

As shown in Fig. 8, the predicted masks align well with anatomical boundaries, suggesting that UniverSeg can generalize beyond microscopy images to a substantially different imaging modality.

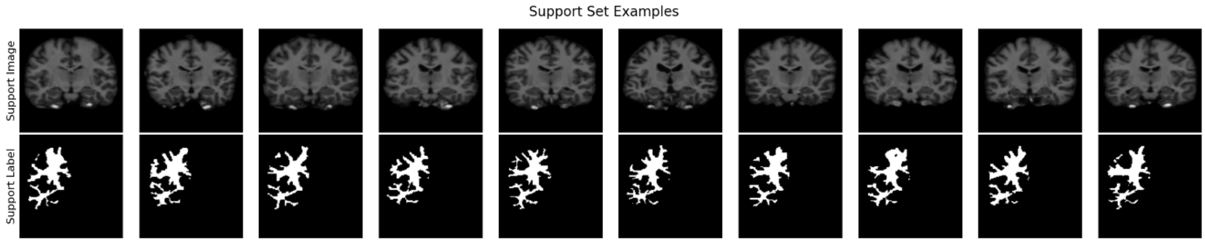

Fig. 8. Qualitative segmentation results on the OASIS brain MRI dataset, demonstrating cross-domain applicability of UniverSeg.

4. Discussion

The experimental results demonstrate that UniverSeg can effectively perform medical image segmentation on previously unseen tasks without any additional training or fine-tuning. By conditioning the segmentation process on a small support set, the model successfully adapts to different targets, such as cytoplasm and nucleus segmentation in WBC images, highlighting its task-agnostic design.

The quantitative results indicate that segmentation performance improves as the support set size increases, particularly in the low-data regime. This behavior suggests that the support set provides crucial contextual information that enables the model to better infer the target structure. Notably, reasonable segmentation performance was achieved even with a small number of support examples, which is advantageous in medical imaging scenarios where annotated data are scarce.

Robust performance under intensity inversion further suggests that UniverSeg relies more on structural relationships encoded through the support set rather than absolute intensity distributions. This property is desirable in real-world clinical environments, where image appearance can vary significantly due to differences in acquisition protocols or imaging devices.

The ensemble experiments reveal that segmentation variability caused by random support set sampling can be effectively reduced by averaging predictions from multiple support configurations. The observed improvement in Dice scores indicates that ensembling enhances prediction stability, particularly when the support set size is limited.

Despite these strengths, the model also exhibits limitations. Performance is dependent on the quality and representativeness of the support set, and suboptimal support examples may lead to degraded segmentation accuracy. In addition, multi-label segmentation was achieved through repeated binary inference rather than a unified multi-class formulation, which may introduce boundary ambiguities in complex scenarios.

5. Conclusion

In this study, we investigated the behavior and performance of a pretrained UniverSeg model for few-shot medical image segmentation. Without any task-specific training, UniverSeg demonstrated strong adaptability across different segmentation targets, support configurations, and imaging domains.

The results suggest that support-conditioned inference provides a practical alternative to conventional retraining-based approaches, particularly in data-limited medical imaging applications. While further improvements are possible through optimized support selection and unified multi-label modeling, UniverSeg represents a promising step toward flexible and reusable medical image segmentation frameworks.